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Refuse container having modular side walls

Abstract

A container including first, second, and third side walls and an open end; a bottom wall coupled with the first, second, and third side walls; and a door pivotably coupled to the container and movable between a first position at which the door is spaced apart from the at least one open end and a second position at which the door closes the at least one open end. The container defines an interior volume of at least ten (10) cubic yards. At least one of the first, second, and third side walls includes a plurality of rectangular side panels releasably fastened together by mechanical fasteners. Each of the plurality of side panels defines a longitudinal axis. The longitudinal axes of each of the plurality of rectangular side panels are parallel with the bottom wall.

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Background/Summary

PRIORITY CLAIM (1) This application is a continuation-in-part of U.S. application Ser. No. 17/844,688, entitled “Refuse Container Having Modular Side Walls,” filed on Jun. 20, 2022, which is hereby relied upon and incorporated herein by reference for all purposes.

TECHNICAL FIELD

(1) Embodiments of the present invention generally relate to the field of material-hauling containers. More particularly, certain embodiments of the present invention relate to improved refuse containers, including but not limited to roll-off dumpsters, having non-welded side wall assemblies. Among other things, various embodiments provide roll-off containers having one or more side walls comprising a plurality of panels that are releasably fastened together.

BACKGROUND

(2) In the waste and materials hauling industry, a variety of containers are used to collect, transport, and/or dump waste, bulk, and liquid materials, among others. Such containers typically are carried on vehicles, such as trucks, or on trailers. The vehicle or trailer is provided with a hoist apparatus to load a container onto and unload the container from the vehicle or trailer, transport the container, and empty the container. Examples of hoist apparatuses include hook hoists, cable hoists, winches, forklifts, and container handlers. The containers loaded and carried by hoist apparatuses may be very heavy, especially when filled. Accordingly, and for example, hoist apparatuses may be rated for a container capacity of 20,000 or 30,000 lbs. Heavier duty hoist apparatuses can be rated for more than 30,000 lbs., in some cases up to 75,000 lbs. or greater.

(3) One type of refuse container is known as a “roll-off container.” Roll-off containers can be used in a variety of demanding waste applications, including scrap collection, construction and remodeling, demolition, and industrial clean-up, among others. Roll-off containers are usually designated by the volume of material they can contain, such as 10, 20, 30, or 40 cubic yards. Roll-off containers are currently available from a number of companies, such as Wastequip, LLC of Charlotte, North Carolina.

(4) Two common types of roll-off containers are rectangular, open top roll-offs and “tub-style” roll offs. The latter style of roll-off container has smooth sides and may be stackable for transporting and storage. FIGS. **1-2** are perspective views of the former style of roll-off container, a rectangular,

open top roll-off container **10**. Container **10** includes a body **12** including walls **14** and a door **16** provided on hinges at one end thereof to facilitate access to the interior volume of container **10**. Container **10** is made out of a suitable metal material, such as steel. Walls **14** can be fabricated from steel, and a floor plate of container **10** (not shown) can be made from 7 gauge steel, for example. Also, walls **14** are reinforced by a plurality of vertical supports **18** which extend between a top rail **20** and a bottom sill **22** of container **10**. Also, container **10** typically includes at least one set of wheels **24** to allow container **10** to be rolled in place and onto and off of a truck frame via a hoist apparatus. In FIGS. **1-2**, two sets of wheels **24** are provided. Container **10** also may include a pair of main rails **26** used to provide support for heavy refuse and to facilitate placement of container **10** on the truck frame.

(5) As shown in FIG. **3**, a vehicle **30** equipped with a cable hoist has a cab **32** and is configured to support a roll-off container **34** on a sub-frame **36** of the cable hoist pivotably connected with vehicle frame. A covering apparatus **38** is provided to extend and retract a cover over the top of container **34**, as is well understood. Vehicle **30** is used for loading, unloading, transporting, and dumping container **34**. For example, the sub-frame **36** of the cable hoist can be elevated and lowered relative to the vehicle frame using hydraulic cylinders, as is also well known. When the sub-frame **36** is elevated to an inclined position, it may serve as a ramp upon which container **34** may be pulled or slid onto and off of the hoist apparatus, for instance using a cable winch system to draw container **34** upward. Container **34** may also be dumped when the sub-frame **36** is in the inclined position. When container **34** is suitably secured on the sub-frame **36**, the sub-frame **36** may be lowered into the position shown in FIG. **3** for transport. Those of skill in the art will appreciate that other types of hoist apparatuses may be used to load, unload, transport, and dump roll-off container **34**, such as a vehicle equipped with a hook hoist. Additional background regarding vehicle mounted hook hoists is provided in U.S. Pat. Nos. 5,542,807; 5,531,559; and 5,088,875, the entire disclosures of which are incorporated herein by reference for all purposes.

(6) The foregoing discussion is intended only to illustrate various aspects of the related art in the field of the invention at the time, and should not be taken as a disavowal of claim scope.

SUMMARY

(7) Some example embodiments comprise apparatus and methods for providing a refuse container with non-welded side and/or bottom walls. In various embodiments, the refuse container may comprise a roll-off container, intermodal container, or any suitable dump body, among others. In various embodiments, a refuse container wall may comprise a plurality of formed sheets overlapped with one another and releasably fastened together (e.g., by bolts). Thus, example embodiments may improve manufacturing efficiency and speed and reduce cost by eliminating many or all welds on wall assemblies, such as with regard to cross-members, top channels, and bottom channels. Various embodiments also provide the ability to manufacture virtually any size (e.g., cubic yardage) of refuse container with standard, and minimal, components. Additionally, various embodiments allow for “spot-fixes” of refuse container walls “in the field,” rather than a full wall or container replacement that would otherwise be required. A damaged wall segment can simply be swapped for a new segment and bolted into place. Embodiments of the invention also provide a roll-off container wall having comparable and/or improved properties (in terms of form and function) relative to existing welded walls.

(8) According to one embodiment, the present invention provides a container comprising a first side wall, a second side wall opposite the first side wall, a front wall coupled with the first and second side walls, a rear wall opposite the front wall and coupled with at least one of the first and second side walls, and a bottom wall coupled with at least the first and second side walls and the front wall. The first side wall, second side wall, front wall, rear wall, and bottom wall together define an interior volume of at least ten (10) cubic yards. At least one of the first side wall, second side wall, front wall, and rear wall comprise a top rail, a bottom rail, a first side panel releasably fastened with the top rail and the bottom rail, and a second side panel releasably fastened with the

top rail, the bottom rail, and the first side panel. The first side panel comprises a first body portion and a first projection having a first flange. The first flange is spaced apart from the first body portion. The second side panel comprises a second body portion and a second projection having a second flange. The second flange spaced apart from the second body portion.
(9) In another embodiment, the present invention provides a container comprising first, second, and third side walls and an open end. The container also comprises a bottom wall coupled with the first, second, and third side walls. Additionally, the container comprises a door pivotably coupled to the container and movable between a first position at which the door is spaced apart from the at least one open end and a second position at which the door closes the at least one open end. The container defines an interior volume of at least ten (10) cubic yards. At least one of the first, second, and third side walls comprises a plurality of vertically-extending side panels releasably fastened together. Each of the plurality of side panels defines a planar body portion and at least one projection from the planar body portion.
(10) According to yet another embodiment, the present invention provides a container comprising vertical side walls and a horizontal bottom wall coupled with the vertical side walls. The vertical side walls and bottom wall together define an interior volume of at least ten (10) cubic yards. At least one vertical side wall comprises a plurality of side panels releasably fastened together. Each of the plurality of side panels defines an inner wall portion and an outer wall portion spaced apart from the inner wall portion. The at least one vertical side wall also comprises a top rail releasably fastened with the inner wall portion of each of the plurality of side panels and a bottom rail releasably fastened with the inner wall portion of each of the plurality of side panels. The outer wall portions of the plurality of side panels extend vertically from the top rail to the bottom rail.
(11) In accordance with one embodiment, the present invention provides a container. The container comprises a first side wall, a second side wall opposite the first side wall, a front wall coupled with the first and second side walls, a rear wall opposite the front wall and coupled with at least one of the first and second side walls, and a bottom wall coupled with at least the first and second side walls and the front wall. The first side wall, second side wall, front wall, rear wall, and bottom wall together define an interior volume of at least ten (10) cubic yards. At least one of the first side wall, second side wall, front wall, and rear wall comprises a top rail and a plurality of side panels releasably fastened together via mechanical fasteners. Each of the plurality of side panels comprises a body portion, at least one projection, and a longitudinal axis. The longitudinal axes of each of the plurality of side panels are parallel with the top rail.
(12) In accordance with another embodiment, the present invention provides a container. The container comprises first, second, and third side walls and an open end; a bottom wall coupled with the first, second, and third side walls; and a door pivotably coupled to the container and movable between a first position at which the door is spaced apart from the at least one open end and a second position at which the door closes the at least one open end. The container defines an interior volume of at least ten (10) cubic yards. At least one of the first, second, and third side walls comprises a plurality of rectangular side panels releasably fastened together by mechanical fasteners. Each of the plurality of side panels defines a longitudinal axis. The longitudinal axes of each of the plurality of rectangular side panels are parallel with the bottom wall.
(13) In yet another embodiment, the present invention provides a container. The container comprises vertical side walls and a horizontal bottom wall coupled with the vertical side walls. The vertical side walls and horizontal bottom wall together define an interior volume of at least ten (10) cubic yards. At least one of the vertical side walls comprises a plurality of side panels releasably fastened together by mechanical fasteners. Each of the plurality of side panels comprises a body portion extending between opposite first and second lateral ends, an upper end, and an opposite lower end. The at least one of the vertical side walls also comprises a top rail releasably fastened together with the upper ends of at least one of the plurality of side panels by mechanical fasteners. The at least one of the vertical side walls further comprises a corner post releasably fastened

together with the first lateral end of at least one of the plurality of side panels by mechanical fasteners. Each body portion of each of the plurality of side panels defines a longitudinal axis that is parallel with the top rail.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Having thus described some example embodiments in general terms, reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:
(2) FIGS. **1-2** are perspective views of a prior art roll-off container;
(3) FIG. **3** is a schematic elevation view of a roll-off container carried on a truck having a hoist apparatus;
(4) FIG. **4**A is a side elevation view of a container according to an embodiment of the present invention;
(5) FIG. **4**B is a plan view of the container of FIG. **4**A;
(6) FIG. **4**C is a front elevation view of the container of FIG. **4**A;
(7) FIG. **4**D is a rear elevation view of the container of FIG. **4**A;
(8) FIG. **5** is a front elevation view of a container side wall assembly according to an embodiment of the present invention;
(9) FIG. **6** is a front perspective view of the container side wall assembly of FIG. **5**;
(10) FIG. **7** is a rear elevation view of the container side wall assembly of FIG. **5**;
(11) FIG. **8** is a rear perspective view of the container side wall assembly of FIG. **5**;
(12) FIG. **9** is a cross-sectional view taken along the line **9-9** in FIG. **5**;
(13) FIG. **10** is a cross-sectional view taken along the line **10-10** in FIG. **5**;
(14) FIG. **11** is a cross-sectional view taken along the line **11-11** in FIG. **5**;
(15) FIG. **12** is a detail view of detail A in FIG. **11**;
(16) FIG. **13** is a front perspective view of a side panel for a container wall assembly in accordance with an embodiment of the present invention;
(17) FIG. **14** is a rear perspective view of the side panel of FIG. **13**;
(18) FIG. **15** is a front elevation view of the side panel of FIG. **13**;
(19) FIG. **16** is a side elevation view of the side panel of FIG. **13**;
(20) FIG. **17** is a top plan view of the side panel of FIG. **13**;
(21) FIG. **18** is a partial exploded perspective view of a container wall assembly and a bottom wall of a container in accordance with an embodiment of the present invention;
(22) FIG. **19** is a partial exploded perspective view of a side panel and top rail of a container wall assembly in accordance with an embodiment of the present invention;
(23) FIG. **20** is a detail cross-sectional view of a container wall assembly in accordance with another embodiment of the present invention;
(24) FIG. **21** is a schematic plan view of three side panels for a container wall assembly in accordance with another embodiment of the present invention;
(25) FIG. **22** is a schematic plan view of three side panels for a container wall assembly in accordance with yet another embodiment of the present invention;
(26) FIG. **23** is a schematic plan view of two side panels for a container wall assembly in accordance with a further embodiment of the present invention;
(27) FIG. **24** is a perspective view of a container in accordance with yet another embodiment of the present invention;
(28) FIG. **25** is a side elevation view of the container of FIG. **24**;
(29) FIG. **26** is a front elevation view of the container of FIG. **24**;
(30) FIG. **27** is a bottom plan view of the container of FIG. **24**;

(31) FIG. **28** is a front perspective view of a side panel of the container of FIG. **24**;
(32) FIG. **29** is a rear perspective view of a side panel of the container of FIG. **24**;
(33) FIG. **30** is a partial exploded perspective view of the container of FIG. **24**, wherein the left side wall assembly and door are not shown;
(34) FIG. **31** is a perspective view of a corner post of the container of FIG. **24**;
(35) FIG. **32** is a perspective view of a floor assembly of the container of FIG. **24**;
(36) FIG. **33** is an exploded perspective view of one of the panel subassemblies of the floor assembly of the container of FIG. **24**;
(37) FIG. **34** is a perspective view of a container in accordance with yet another embodiment of the present invention; and
(38) FIG. **35** is a partial exploded perspective view of the container of FIG. **34**, wherein the left side wall assembly is not shown.
(39) Repeat use of reference characters in the present specification and drawings is intended to represent same or analogous features or elements of embodiments of the present invention.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

(40) Reference will now be made in detail to presently preferred embodiments of the invention, one or more examples of which are illustrated in the accompanying drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that modifications and variations can be made in the present invention without departing from the scope or spirit thereof. For instance, features illustrated or described as part of one embodiment may be used on another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.
(41) As used herein, terms referring to a direction or a position relative to the orientation of a container, such as but not limited to “vertical,” “horizontal,” “upper,” “lower,” “front,” or “rear,” refer to directions and relative positions with respect to the container's orientation in its normal intended operation, as indicated in the Figures herein. Thus, for instance, the terms “vertical” and “upper” refer to the vertical direction and relative upper position in the perspectives of the Figures and should be understood in that context, even with respect to an apparatus that may be disposed in a different orientation. The term “substantially,” as used herein, should be interpreted as “nearly” or “close to”, such as to account for design and manufacturing tolerances of the apparatus.
(42) Embodiments of the present invention relate to improved systems and methods for providing modular refuse container side and/or bottom wall assemblies. As noted above, in various embodiments, a container wall assembly may comprise a plurality of overlapping sheets (e.g., of metal) that are releasably fastened together. In some embodiments, the wall assembly may have a corrugated shape. In some embodiments, the side panels also may be interlocking without the use of fasteners.
(43) Although some preferred embodiments are discussed below in the context of rectangular, open-topped roll-off containers, those of skill in the art will appreciate that the present invention is not so limited. In particular, it is contemplated that embodiments of the present invention may be used with any suitable waste, refuse, or payload container, such as but not limited to tub-style roll-off containers, intermodal containers, dump bodies, front-end load containers, and rear-end load containers. Further, it is contemplated that various embodiments may be used with any one or all of the four container side walls, and/or the top or bottom wall of a container. In certain embodiments, the modular wall assemblies may comprise only part of a wall, rather than the entire wall. Also, although certain embodiments are depicted with a vertically oriented side panel, in other embodiments the side panels may be disposed at an angle to vertical and/or arranged horizontally.
(44) Turning now to the Figures, certain details of a container constructed in accordance with an embodiment of the present invention are described below with reference to FIGS. **4**A-**4**D and **5-18**. In general, a container **100** may be a rectangular open-topped roll-off container comprising a

bottom wall **102**, two side wall assemblies **104** in facing opposition, a front wall **106**, and a door **108**. Bottom wall **102**, side wall assemblies **104**, front wall **106**, and door **108** are coupled together to define an interior volume therein. In various embodiments, the interior volume may be at least ten (10) cubic yards, though other interior volumes may be defined in other embodiments. Door **108** is coupled with one of side wall assemblies **104** in this embodiment via hinges **110**, which permit door **108** to open and close the interior volume of container **100**. A latching device **112** may be coupled to one of side wall assemblies **104** to latch door **108** in a closed position as is understood. As those of skill in the art will appreciate, a door **108** may not be provided in all embodiments, and instead a fixed rear wall could be coupled between side wall assemblies **104**.
(45) In the past, the side walls of a container like container **100** typically would be formed from a rectangular metal sheet, for example formed of 12 gauge or 7 gauge steel. In the illustrated embodiment, however, each side wall assembly **104** comprises a plurality of side panels **114**. The number of side panels **114** provided depends on the size of each side panel, the size or volume and shape of container **100**, the orientation of side panels **114**, and the particular side of container **100** at issue. Where, as shown, a side wall assembly is provided on the long sides of an open-topped roll-off container and side panels **114** are oriented vertically, in one embodiment ten (10) such panels may be provided. In other embodiments and for other sides of a container, different numbers of panels **114** may be used. In that regard, the embodiment of container **100** shown in the figures comprises side wall assemblies **104** along its two long sides, with front wall **106** and door **108** being analogous to those used in conventional roll-off containers. It will be appreciated, however, that front wall **106** (or a fixed rear wall) may also or alternatively comprise side wall assemblies as described herein in other embodiments. Further, in certain other embodiments, bottom wall **102** and/or a door **108** may also comprise a plurality of side panels **114** or other modular panels, as described herein.
(46) Side wall assemblies **104** also comprise top rails **116** that are releasably fastened (as described in more detail below) with and extend horizontally along an upper edge of side panels **114**. In various embodiments, top rails **116** may comprise a length of rectangular metal tubing. The lower edge of side panels **114** may be releasably fastened with bottom rails **118**, which are also preferably coupled with bottom wall **102**. Corner posts **120**, where provided, may be coupled with the lateral edges of a panel **114** and extend between top rails **116** and bottom wall **102**. As described in more detail below, in some embodiments projections on side panels **114** together define a plurality of vertical supports **122** that extend vertically between top rails **116** and bottom rails **118**. As will be appreciated, such vertical supports **122** may stiffen walls **104** and increase the bending strength thereof.
(47) In this embodiment, front wall **106** may comprise a rectangular metal sheet **124** that extends vertically between a top rail **126** and bottom rails **128** coupled with respective upper and lower edges thereof. Metal sheet **124** also extends laterally between corner posts **120**. Bottom rails **128** may also be coupled with bottom wall **102**. Corner wraps **130** may be provided between top rails **116** and top rail **126** for added strength. Also, one or more vertical supports **132** may extend vertically between top rail **126** and bottom rails **128** to stiffen or otherwise provide support to front wall **106**.
(48) As with container **10** described above, in various embodiments container **100** may also comprise at least two wheels **134**. In the illustrated embodiment, four such wheels **134** are provided. A substructure of container **100** may comprise a pair of longitudinal rails **136** which extend generally along the length thereof and which may provide support for bottom wall **102** and facilitate loading, unloading, and transport of container **100** on a hoist frame.
(49) An example container side wall assembly **104** is described in more detail with reference to FIGS. **5-18**. In the illustrated embodiment, side wall assembly may be about 8′ tall and 22′ long, though as noted above the dimensions of side wall assembly **104** will vary depending on the type of container and side wall at issue. Also as noted above, side wall assembly **104** comprises a plurality

of side panels **114** that are coupled with top rail **116** and bottom rail **118** via a non-welded connection, such as mechanical fasteners. In various embodiments, suitable fasteners include, but are not limited to, bolts, screws, nails, and other mechanical fasteners. Among many possibilities, suitable bolts may include hex bolts, huck bolts, carriage bolts, and shoulder bolts. In one embodiment, the fasteners may comprise 7/16″ bolts and corresponding weld nuts. Nonetheless, those of skill in the art will appreciate that fasteners other than bolts may be suitable in various embodiments. Also, those of skill in the art will appreciate that the type, number, and grade of fasteners used may vary in various applications and for various intended strengths of the container. For higher strengths, the diameter of fasteners may be increased and the number of fasteners used may increase.

(50) In one embodiment, side panels **114** comprise a sheet of suitable strength for refuse disposal environments (e.g., 7 or 12 gauge steel), though other suitable materials may be used. Thus, side panels **114** may be planar in shape. In some embodiments, each side panel **114** may be formed from a single unitary sheet, though this is not required in all embodiments. Side panels **114** also may not comprise a sheet in some embodiments, and may instead define a tubular or double-walled structure. Also, although side panels **114** are illustrated as being generally rectangular, side panels **114** may define any suitable shape in other embodiments.

(51) In any event, as shown, side panels **114** may be generally planar and have a body portion **137** comprising a first surface **138**, which may face the exterior of a container upon assembly, and an opposite second surface **139**, which may face the interior of a container upon assembly. Side panels **114** in this embodiment also may comprise a first projection **140** and a second projection **142**. In other words, in some embodiments, side panels **114** comprise an inner wall portion (e.g., body portion **137**) and an outer wall portion (e.g., one or more projections **140**, **142**) spaced apart from the inner wall portion.

(52) First and second projections **140** and **142** respectively define flanges **144**, **146** in this embodiment. Flanges **144** and **146** are spaced apart from body portion **137**. In one embodiment, side panels **114** may be stamped and flanges **144**, **146** may be defined during the stamping process. In other embodiments, flanges **144**, **146** or other projections (as described elsewhere herein) may be defined by folding or bending panels **114** during manufacturing. Further, in some embodiments, projections **140**, **142** may not be unitary with body portions **137**. Also, in some embodiments, a side panel **114** may comprise only a single projection **140** or **142** or may comprise more than two projections **140** or **142**.

(53) As noted above, side panels **114** may be releasably fastened to top rail **116** and bottom rail **118**. In this regard, and for example, body portion **137** of each side panel **114** may define a lower end **148** and an upper end **150**. Each end **148**, **150** defines a plurality of apertures **152** configured to receive suitable fasteners therethrough. In one embodiment, apertures **152** may be about 0.5″ in diameter, though again the size may vary. In this embodiment, there are four (4) apertures **152** on each end **148**, **150**. As shown, e.g., in FIGS. **7-8** and **19**, each of top rail **116** and bottom rail **118** also define a plurality of apertures **153**. The apertures are positioned so that, when the first surface **138** of body portion **137** is disposed against a respective rear surface **154**, **156** of top rail **116** and bottom rail **118**, they align with apertures **152**. Thereby, each side panel **114** may be releasably fastened to top rail **116** and bottom rail **118** via the fasteners.

(54) Thus, the height dimension of body portion **137** may be selected to correspond to the vertical distance between top rail **116** and bottom rail **118**. As shown in FIG. **7**, for example, the top edge of side panels **114** may be aligned with the top surface of top rail **116**, and the bottom edge of side panels **114** may be aligned with the bottom surface of bottom rail **118**. In other embodiments, however, side panels **114** may be releasably fastened to top rail **116** and bottom rail **118** in another manner, and as such the height dimension of body portion **137** may vary in other embodiments. For instance, either or both of lower end **148** and upper end **150** may be partially or entirely bent so that it can be attached to an upper or lower surface of bottom rail **118** and/or top rail **116**, rather than to

the rear surfaces **156**, **154** thereof.

(55) In certain embodiments, each side panel **114** may partially overlap with another side panel **114** when side wall assembly **104** is assembled. The overlapping feature in this embodiment helps maintain both the structural integrity and the watertight properties of conventional containers while eliminating welded seams. For example, either or both projection **140**, **142** of each side panel **114** may overlap with another projection **140**, **142** of an adjacent side panel **114**. More particularly, the locations of the apertures defined in top rail **116** and bottom rail **118** may be selected so that, when side panels **114** are fastened to top rail **116** and bottom rail **118**, their relative positions cause a projection **140**, **142** of one side panel to overlap with a projection **140**, **142** of an adjacent side panel. In other embodiments described in more detail below, side panels **114** may overlap at locations on their respective body portions, rather than at their respective projections. Also, side panels **114** may not overlap at all in still other embodiments.

(56) Additionally, each side panel **114** is releasably fastened to its adjacent side panel(s) **114**. In the illustrated embodiment, the flange(s) **144**, **146** of each side panel **114** preferably define a plurality of apertures **158**. In one embodiment, apertures **158** may be centered on flanges **144**, **146** and spaced every 12″, though the number of apertures **158** used may vary in other embodiments. As shown, each side panel flange **144**, **146** may define eight (8) apertures **158**. Apertures **158** preferably are located in each flange **144**, **146** so that, when two adjacent side panels **114** are assembled to top rail **116** and bottom rail **118**, the apertures **158** in the respective flange **144** or **146** of each side panel **114** will be in alignment (see, e.g., FIGS. **11-12**). Accordingly, fasteners (e.g., bolts) may be inserted through the aligned apertures **158** and used to secure side panels **114** together.

(57) As best seen in FIG. **11**, in a given container wall assembly **104**, side panels **114** may be assembled so that their respective projections **140**, **142** alternate between being in front of (or exterior of) the projections **140**, **142** of adjacent side panels and being behind (or interior of) the projections **140**, **142** of adjacent side panels. For instance, the side panel **114** that is completely shown in FIG. **11** has its projections **140**, **142** disposed in front of (or exterior of) the projections **142**, **140** of each of its adjacent side panels **114** (which are only partially shown in FIG. **11**). However, this particular configuration is not required in all embodiments. Each panel **114** could have its projection **142** disposed behind (or interior of) a projection **140** of one adjacent panel and its projection **140** in front of (or exterior of) a projection **142** of the other adjacent panel, or vice versa. Depending on the material from which each side panel **114** is made, side panel **114** may be flexible enough to allow assembly in any of these configurations. In other embodiments, such as those discussed with regard to FIGS. **21-22** below, side panels **114** may be dimensioned to be assembled in one particular configuration or orientation.

(58) In the illustrated embodiment, flanges **144** and **146** are co-planar and offset from the plane in which body portion **137** lies. Likewise, where the assembly configuration illustrated in FIG. **11** is used, the body portions **137** of adjacent panels will be slightly offset and will not be co-planar. However, this is not required in all embodiments. For instance, flanges **144** and **146** may extend in parallel, but offset planes. Likewise, in some configurations, body portions **137** of adjacent side panels **114** may extend in the same plane, or each body portion **137** may extend in a different plane.

(59) When projections **140**, **142** of adjacent side panels **114** are fastened together in this embodiment, they together define the plurality of vertical supports **122** mentioned above. These vertical supports **122** may be analogous in function to the vertical supports **18** of conventional open-topped roll-off containers in some embodiments. When viewed in plan or cross-section, vertical supports **122** may have a polygonal shape in this embodiment, but vertical supports may have any suitable shape in other embodiments.

(60) Vertical supports **122** preferably extend from the top rail **116** to the bottom rail **118**. More specifically, in one embodiment, the vertical dimension of projections **140**, **142** is preferably selected to correspond to the vertical distance between the top surface of bottom rail **118** and the

bottom surface of top rail **116**, so that, following assembly, the bottom edges of flanges **144**, **146** engage the top surface of bottom rail **118** and the top edges of flanges **144**, **146** engage the bottom surface of top rail **116**. Thus, projections **140**, **142** may be shorter in height than body portion **137** in some embodiments.

(61) Depending on the application for which container **100** will be used, in various embodiments, a suitable gasket material (e.g., rubber gasket **160** in FIG. **20**) may be disposed between projections **140**, **142** of adjacent side panels **114** to help maintain watertightness. Likewise, in various embodiments, a suitable gasket material also may be disposed between the portions of side panels **114** that are in engagement with top rail. For instance, gaskets could be disposed between body portion **137** and the rear surfaces **154**, **156** of top rail **116** and/or bottom rail **118**. Gaskets also could be disposed between the bottom surface of top rail **116** and the top edges of flanges **144**, **146** and/or between the top surface of bottom rail **118** and the bottom edges of flanges **144**, **146**. However, gasketing is not required in all embodiments.

(62) With reference now in particular to FIG. **18**, as noted above, bottom rail **118** is coupled with a bottom wall **102** of container **100**. In some embodiments, one or more side panels **114** also are coupled with bottom wall **102**. For example, bottom wall **102** optionally may define a plurality of locating features (e.g., slots **162**) and each applicable side panel **114** optionally may define a plurality of correspondingly-sized locating features (e.g., tabs **164**). Tabs **164** may extend from lower end **148** of each applicable side panel **114**, and may be any suitable shape. As shown, tabs **164** are rectangular in shape. Slots **162** are located in bottom wall **102** and configured to receive tabs **164** and help position side panels **114** during assembly of side wall assembly **104**. This or another suitable coupling between side panels **114** and bottom wall **102** also may add rigidity to container **100**.

(63) FIG. **19** is an exploded view of a side wall **114** and top rail **116** in accordance with another embodiment. Here, top rail **116** and projections **140**, **142** may define similar locating features. For instance, projections **140**, **142** may each comprise one or more tabs **166** extending vertically from upper edges thereof. A bottom surface **168** of top rail **116** may define correspondingly sized slots **170** that are located and configured to receive tabs **166** during assembly of side wall assembly **104**. In various embodiments, tabs also could be disposed on the lower edges of projections **140**, **142** and configured to mate with corresponding slots defined in a top surface of bottom rail **118**. Any such tabs may be provided in addition to or in the alternative to the tabs described with reference to FIG. **18** in various embodiments.

(64) FIG. **20** is a detail cross-sectional view of a container wall assembly in accordance with another embodiment of the present invention. This figure is analogous to the view of FIG. **12**, but in this embodiment, the container wall assembly comprises one or more interior wall plates **172**. As shown, wall plates **172** optionally may be provided between adjacent side panels **114** at locations opposite the overlapping flanges **144**, **146**. Thus, wall plates **172** may be located on the interior wall (e.g., second surface **139**) side of wall assembly **104**. Wall plates **172** may be provided, for example, to create a more uniform interior wall in container **100** and to increase the watertightness of container **100**. Wall plates **172**, where provided, may also increase the strength and rigidity of container **100** and vertical supports **122**. Wall plate **172** may be rectangular in shape and sized to fill and/or cover a recess **174** in the interior wall created by projections **140**, **142**. In the illustrated embodiment, wall plate **172** is wider than recess **174** so that its lateral edges abut second surface **139** of side panels **114**. In other embodiments, wall plate **172** could be dimensioned to be received with recess **174** in other embodiments such that the interior-facing side of wall assembly **104** defines a continuous, smooth surface. Wall plate **172** may be coupled with side panels **114** via a bolt **176** that is received through apertures **158** defined in flanges **144**, **146** and a nut **178** that is received on bolt **172** and tightened against wall plate **172**.

(65) It is contemplated that, in various embodiments, side panels of a container wall assembly may have cross-sectional shapes that differ from the cross-sectional shape of side panels **114** described

above. Each side panel may be the same as its adjacent side panel(s) in some embodiments, but that is not required. Alternating side panels, or portions thereof, could have different dimensions or shapes, including differing thicknesses and/or widths.

(66) In that regard, and turning now to FIG. **21**, a portion of a side wall assembly comprising three side panels **200** is schematically illustrated in plan view. In this embodiment, each side panel **200** defines a body portion **202** and a projection **204**. Unlike side panels **114** described above, here side panels **200** are releasably fastened together at their respective body portions **202**, rather than at their respective projections **204**. Body portions **202** define apertures **206** in which bolts **208** are received to releasably fasten side panels **200** to one another. Thus, in this embodiment, each side panel comprises its own vertically extending support, unlike the vertical supports **122** that were formed by the assembly of the projections of adjacent side panels.

(67) Also unlike side panels **114**, in this embodiment, projections **204** are semicircular when viewed in plan or cross-section. Further, in this embodiment, each side panel **200** is not symmetrical when viewed in plan. Rather, the left-hand side of the body portion **202** is disposed in a plane that is parallel with but offset from (e.g., behind or interior of) the plane in which the right-hand side of body portion **202** lies.

(68) In the embodiment of FIG. **22**, side panels **210** define projections **212** that are rectangular in shape when viewed in plan or cross-section. Also, in this embodiment, the width of each projection **210** is greater than the width of body portions **214**. In this embodiment, unlike the embodiment of FIG. **21**, the plane in which the left-hand side of body portion **214** lies is in front of (or exterior of) the plane in which the right-hand side of body portion **214** lies.

(69) FIG. **23** illustrates an embodiment including two side panels **250**. In this embodiment, side panels **250** are releasably fastened together but do not overlap one another. In particular, side panels **250** comprise a body portion **252** and a projection **254** therefrom. Body portion **252** also defines two lateral flanges **256**. Each flange **256** defines an aperture **258** configured to receive a suitable fastener, such as a bolt **208**. When side panels **250** are disposed side-by-side, flanges **256** of adjacent panels are in engagement, and their respective apertures **258** are in alignment. In the illustrated embodiment, flanges **256** extend toward the interior volume of a container, opposite the direction of projections **254**. In other embodiments, flanges **256** could extend in the same direction as projections **254**.

(70) A container **300** constructed in accordance with yet another embodiment of the invention is shown in FIGS. **24-33**. In this regard, FIGS. **24-27** are perspective, side elevation, rear elevation, and bottom plan views of container **300**, respectively. FIGS. **28** and **29** are front and rear perspective views of a side panel of container **300**. FIG. **30** is a partial exploded perspective view of container **300**, wherein the left side wall assembly and door are not shown. FIG. **31** is a perspective view of a corner post of container **300**. FIG. **32** is a perspective view of a floor assembly of container **300**, and FIG. **33** is an exploded perspective view of one of the panel subassemblies of the floor assembly of container **300**.

(71) In this embodiment, container **300** is a rectangular open-topped roll-off container comprising a floor assembly **302**, two side wall assemblies **304** in facing opposition, a front wall **306**, and a door **308**. Floor assembly **302**, side wall assemblies **304**, front wall **306**, and door **308** are coupled together to define an interior volume therein. In various embodiments, the interior volume may be at least ten (10) cubic yards, though other volumes may be defined in other embodiments. Also, in various embodiments, each side wall assembly **304** may be about 8′ tall and 22′ long, though again these dimensions will vary depending on the type of container and side wall at issue.

(72) Here, each side wall assembly **304** of container **300** comprises a corner post **310**, a plurality of side panels **312**, and a top rail **314**. The number of side panels **312** depends on the size of each side panel, the size or volume and shape of container **300**, the orientation of side panels **312**, and the particular side of container **300** at issue. Where, as shown, a side wall assembly is provided on the long sides of an open-topped roll-off container, in one embodiment nine (9) such panels may be

provided in each side wall assembly **304**. As described in more detail below, in this embodiment, side panels **312** are oriented horizontally (e.g., with a longitudinal axis **315** of each side panel **312** being parallel with an axis defined along the long dimension of container **300**), rather than vertically or at another angle. Of course, in other embodiments and for other sides of a container, different numbers and configurations of panels **312** may be used. Again, in the embodiment of container **300**, a plurality of side panels **312** also could be used for any of front wall **306**, door **308**, and/or floor assembly **302**.

(73) Door **308** is coupled with one of side wall assemblies **304** in this embodiment via hinges **316** coupled with one of the corner posts **310**. Hinges **316** permit door **308** to open and close the interior volume of container **300**. As those of skill in the art will appreciate, door **308** may not be provided in all embodiments, and instead a fixed rear wall could be coupled between side wall assemblies **304**.

(74) Top rails **314** are coupled with corner posts **310** and extend horizontally along an upper edge of three side panels **312** in this embodiment. As shown, top rails **314** comprise a length of rectangular metal tubing. In this embodiment, bottom rails analogous to bottom rails **118** of container **100** above are not provided, but they can be in other embodiments. Corner posts **310** are coupled with the lateral edges of several side panels **312** and extend between top rails **314** and floor assembly **302**. In various embodiments, corner posts **310** can be coupled with floor assembly **302** via any suitable method, including mechanical fasteners as described herein or via welding, either alone or in combination with locating features such as those described above in connection with FIG. **18**. In a mechanically-fastened application, various embodiments may also include a mounting bracket and/or securing gusset.

(75) In this embodiment, front wall **306** comprises a rectangular metal sheet **318** that extends vertically between a top rail **320** and floor assembly **302**. Again, in some embodiments, a bottom rail analogous to bottom rail **128** may be provided. In various embodiments, front wall **306** can be coupled with floor assembly **302** in the same manner as corner posts **310**, described in the preceding paragraph. Top rail **320** is coupled at each of its ends with one of top rails **314**. Metal sheet **318** also extends laterally between the lateral edges of several side panels **312** of each side wall assembly **304**. As best seen in FIGS. **26** and **30**, the lateral edges of metal sheet **318** may define a shape that corresponds to the shape of the panels **312** that form each side wall assembly **304**. For instance, rather than defining a straight vertical edge, the edges of metal sheet **318** may define a plurality of notches **322** so that the edges of sheet **318** correspond to the profile of each side wall assembly **304**. Alternatively, in some embodiments, metal sheet **318** could define one or more flanges that project from the edges shown in FIG. **26** perpendicularly from metal sheet **318** toward the interior volume of container **300**. Such flanges may be shaped to define a surface that corresponds to an interior surface of the connected side panels **312** of side wall assembly **304** so that the side panels **312** and flange can be coupled together (e.g., releasably fastened). In some embodiments, metal sheet **318** may be formed from a stamping process, and flanges, if provided, may be bent from metal sheet **318**. Of course, other methods of forming flanges, if provided, and of connecting metal sheet **318** and side panels **312** are within the scope of the present invention. In various embodiments, one or more horizontal and/or vertical support members **324** can extend along metal sheet **318** to stiffen or otherwise provide support to front wall **306**.

(76) Referring now also to FIG. **31**, corner posts **310** in this embodiment comprise a metal channel having a body portion **326** and a pair of flanges **328** that project perpendicularly from body portion **326**. In other embodiments, corner posts **310** could comprise rectangular metal tubing or the like. Here, during assembly, posts **310** are arranged so that flanges **328** face the interior volume of container **300**. Similar to the edges of metal sheet **318**, described above, the distal edges of each flange **328** define a shape that corresponds to the shape of the panels **312** that form the side wall assembly **304** that will be connected with a given post **310**. For instance, rather than defining a straight vertical edge, the edges of each flange **328** define a plurality of notches **330** so that the

flanges mate flush with a side wall assembly **304** once assembled together. Alternatively, in some embodiments, corner posts **310** could define one or more interior surfaces that extend partially or completely between flanges **328** and that define apertures that align with corresponding apertures in the panels **312** to which they will be connected. In some embodiments, corner posts **310** need not have notches at all, and they may instead define a straight length of metal tubing or metal channel that can be coupled (e.g., releasably fastened) with one or more side panels **312**.

(77) Although not shown in this embodiment, in various embodiments container **300** may also comprise wheels analogous to wheels **134**, described above. Also, a substructure of container **300** can comprise a pair of longitudinal rails analogous to rails **136**, described above.

(78) Side wall assemblies **304** in this embodiment are described with particular reference to FIGS. **24** and **28-30**. As noted above, each side wall assembly **304** comprises a plurality of side panels **312** that, in various embodiments, can be coupled together via a non-welded connection, such as mechanical fasteners. However, in certain embodiments, welding could be used to fasten side panels together alone or in addition to the use of mechanical fasteners. In the configuration shown in these figures, certain side panels **312** are coupled with top rail **314**, certain are coupled with corner post **310**, certain are coupled with floor assembly **302**, and certain are coupled with metal sheet **318** of front wall **306**. In various embodiments, the side panels **312** that are to be coupled to the floor assembly **302** and to front wall **306** can be so coupled in the same manner as corner posts **310**, described above. In this regard, a side panel **312** as shown in FIGS. **28-29** can define a plurality of apertures at various positions around its perimeter that, during assembly, can be aligned with correspondingly-defined apertures in other side panel(s) **312**, corner posts **310**, and/or top rail **314** (not shown, but analogous to those described above in connection with container **100**) to releasably fasten these various components to one another. As those of skill in the art will appreciate, the number and placement of apertures in a given side panel **312** will depend on its location within a side wall assembly **304**, and each side panel **312** may not have the same number and/or configuration of apertures as the other side panels **312**. Thus, apertures in side panel **312** are not shown in this embodiment, but they can be analogous to the apertures **152** of side panels **114** shown above, and the apertures in corner posts **310** and/or top rail **314** can be analogous to apertures **153**, described above. In various embodiments, suitable fasteners include, but are not limited to, bolts screws, nails, and other mechanical fasteners such as those described in connection with container **100** above, and of course the type, number, and grade of fasteners used will vary, as needed or desired, in various embodiments.

(79) In this embodiment, side panels **312** comprise a metal sheet of suitable strength for refuse disposal applications (e.g., 7 or 12 gauge steel), though other suitable materials may be used. Side panels **312** can be generally planar in shape and formed from a single unitary sheet, though this is not required in all embodiments. As with side panels **114** above, side panels **312** also could comprise a tubular or double-walled structure in other embodiments. Although side panels **312** are illustrated as being generally rectangular, side panels **312** can define any suitable shape in other embodiments.

(80) As shown, side panels **312** define a body portion **334** comprising a first surface **336**, which may face the exterior of a container upon assembly, and an opposite second surface **338**, which may face the interior of a container upon assembly. Of course, the orientation of surfaces **336**, **338** may be reversed in other embodiments. Side panels **312** in this embodiment also define a first projection **340** and a second projection **342**. Side panels **312** can, in some embodiments, comprise an inner wall portion (e.g., one or more projections **340**, **342**) and an outer wall portion (e.g., body portion **334**). Thus, side panels **312** in this embodiment are analogous in certain respects to side panels **114** described above, although the body portion **334** is different in dimensions than body portion **137**, projections **340**, **342** are different in dimension from projections **140**, **142**, and projections **340**, **340** do not define flanges analogous to flanges **144**, **146** of side panels **114**. In some embodiments, side panels **312** can be formed of stamped metal, and projections **340**, **342** may

be defined by folding or bending panels **312** during manufacturing. It is also not required that projections **340**, **342** be unitary with body portion **334** in all embodiments. In various embodiments, a side panel **312** can define only a single projection **340** or **342** or more than two projections **340**, **342**.

(81) Also in this embodiment, side panels **312** define an upper end **344**, a lower end **346**, a first lateral end **348**, and an opposite second lateral end **350**. As noted above, in various embodiments, various side panels **312** can be releasably fastened to one another, to top rail **314**, to corner post **310**, to floor assembly **302**, and/or to front wall **306**. For example, with reference to the exploded view of FIG. **30**, in which the second surfaces **338** of nine (9) side panels **312** are visible, three (3) side panels **312***a*, **312***b*, and **312***c* together form a first, or top, row of side panels. These side panels can be releasably fastened with top rail **314** via apertures defined along their respective upper ends **344** that are configured to receive suitable fasteners therethrough. Top rail **314** can define a plurality of apertures that are positioned so that, when the first surfaces **336** are disposed against the interior-facing surface **352** of top rail **314**, they align with the apertures defined in the upper ends **344** of side panels **312**. Thereby, each side panel **312** can be releasably fastened to top rail **314**. Where a bottom rail is provided, other side panels **312** could similarly be attached to the bottom rail. Additionally, as shown, lateral end **350** of side panel **312***a* is releasably fastened with metal sheet **318** of front wall **306**, and lateral end **348** of side panel **312***c* is releasably fastened with corner post **310**.

(82) Lateral end **348** of side panel **312***a* is releasably fastened with lateral end **350** of side panel **312***b*, and lateral end **350** of side panel **312***c* is releasably fastened with lateral end **348** of side panel **312***b*. Specifically, and for example, side panel **312***a* may define one or more apertures defined on lateral end **348**, and side panel **312***b* may define one or more apertures defined on lateral end **350**. These respective apertures can be positioned so that when panels **312***a* and **312***b* are arranged side by side with lateral end **348** of side panel **312***a* disposed either in front of or behind lateral end **350** of side panel **312***b* (relative to the exterior of container **300**), the respective apertures are in alignment and suitable fasteners can be used in the apertures to couple side panel **312***a* with side panel **312***b*. In other words, in this embodiment, side panels **312***a* and **312***b* can overlap by a predetermined amount in the lateral direction. Lateral end **350** of side panel **312***c* can be releasably fastened with lateral end **348** of side panel **312***b* in a similar, partially overlapping fashion. Thus, side panels **312***b* and **312***c* also can overlap by a predetermined amount in the lateral direction. In this embodiment, the lateral distance between corner post **312** and metal sheet **318** can be less than the sum of the lengths of panels **312***a*, **312***b*, and **312***c*.

(83) Also, three (3) side panels **312***d*, **312***e*, and **312***f* together form a second, or middle, row of side panels in this embodiment. These side panels can be releasably fastened with each other in partially overlapping fashion, with metal sheet **318** of front wall **306**, and with corner post **310** in the same manner described above. The respective upper ends **344** of each of these side panels **312***d*, **312***e*, and **312***f* are releasably fastened with respective lower ends **346** of each of the side panels **312***a*, **312***b*, and **312***c*. In this regard, upper end **344** of side panel **312***d* can partially overlap with (either in front of or behind relative to the exterior of container **300**) lower end **346** of side panel **312***a*, such that one or more apertures defined in upper end **344** of side panel **312***d* come into alignment with one or more corresponding apertures defined in lower end **346** of side panel **312***d*. Upper ends **344** of side panels **312***e* and **312***f* are releasably fastened with respective lower ends **346** of side panels **312***b* and **312***c* in the same, partially overlapping manner

(84) Similarly, three (3) side panels **312***g*, **312***h*, and **312***i* together form a third, or bottom, row of side panels in this embodiment. These side panels can be releasably fastened with each other in partially overlapping fashion, with metal sheet **318** of front wall **306**, and with corner post **310** in the same manner described above. The respective upper ends **344** of each of these side panels **312***g*, **312***h*, and **312***i* are releasably fastened with respective lower ends **346** of each of the side panels **312***d*, **312***e*, and **312***f*, also in the same, partially overlapping manner described above. The

respective lower ends **346** of each of these side panels **312***g*, **312***h*, and **312***i* also can be releasably fastened with bottom wall **302** in the same manner as described above in connection with FIG. **18** or via welding.

(85) As described above, in coupling side panels **312***a*-**312***i* to one another, each side panel **312** can partially overlap with (i.e., is partially in front of or behind) at least two sides or ends of another side panel **312**. As a result, certain side panels **312** may be closer to the exterior of container **300** than other side panels **312**, and certain side panels **312** may be closer to the interior of container **300** than other side panels **312**. Many different configurations for releasably fastening a plurality of side panels **312** together are possible and within the scope of the present disclosure. In some embodiments, the side panels **312** in a side wall assembly **304** can alternate between being in front or behind other side panels **312** in a pattern analogous to that of the black and white squares on a checkerboard.

(86) One example of this is best seen in FIG. **30**, wherein a portion of the second surface **338** of panel **312***a* is in front of (e.g., more exterior of) and disposed against portions of the first surfaces **340** of panels **312***b* and **312***d*. Likewise, a portion of the second surface **338** of panel **312***e* is in front of (e.g., more exterior of) and disposed against portions of the first surfaces **340** of panels **312***b*, **312***d*, **312***f*, and **312***h*. A portion of the second surface **338** of panel **312***c* is in front of (e.g., more exterior of) and disposed against portions of the first surfaces **340** of panels **312***b* and **312***f*. A portion of the second surface **338** of panel **312***g* is in front of (e.g., more exterior of) and disposed against portions of the first surfaces **340** of panels **312***d* and **312***h*. And a portion of the second surface **338** of panel **312***i* is in front of (e.g., more exterior of) and disposed against portions of the first surfaces **340** of panels **312***f* and **312***h*. In all cases, apertures defined in the ends of various side panels **312** are brought into alignment with corresponding apertures defined in the ends of the side panels **312** that they engage. Thus, in this embodiment, panels **312***a*, **312***c*, **312***e*, **312***g*, and **312***i* are in front of panels **312***b*, **312***d*, **312***f*, and **312***h* when viewed from the exterior of container **300**.

(87) Of course, in another embodiment, the opposite arrangement could be provided. In yet other embodiments, when viewed from the exterior of container **300**, panel **312***a* could be disposed in front of panels **312***b* and **312***d*; panels **312***b* and **312***d* could be disposed in front of panels **312***c*, **312***e*, and **312***g*; panels **312***c*, **312***e*, and **312***g* could be disposed in front of panels **312***f* and **312***h*; and panels **312***f* and **312***h* could be disposed in front of panel **312***i*, which could be the innermost panel. This arrangement could of course be done with any of side panels **312***c*, **312***g*, and/or **312***i* as the outermost panel. In other arrangements, panel **312***e* could be either the outermost panel or the innermost panel. All such other arrangements are contemplated.

(88) Moreover, in some embodiments in which a side wall assembly **304** includes a plurality of side panels **312**, some or all of the various side panels **312** can be non-overlapping. In other words, and for example, edges of some or all of the side panels **312** could engage the edges of adjacent side panels **312** without being in front of or behind such edges. In this case, vertical and/or horizontal posts and/or cross-members (e.g., analogous to corner post **310** and support members **324**) could be provided on the interior or exterior side of side wall assembly **304** such that they are disposed over the interface between two or more side panel **312** edges. These posts and/or cross members can be provided with apertures that are aligned with corresponding apertures defined in the various ends of side panels **312** once disposed in the proper position for assembly. Thereby, suitable fasteners could be used to releasably secure these components together. In other embodiments, adjacent edges of side panels **312** could define inwardly- or outwardly-facing flanges (e.g., analogous to flanges **256**, described above) that allow the various side panels **312** to be releasably fastened together.

(89) Next, FIG. **32** illustrates an embodiment of floor assembly **302**. In various embodiments, floor assembly **302** is modular and comprises a plurality of panel subassemblies **354** that are coupled (e.g., releasably fastened) together. One example panel subassembly **354** is illustrated in the exploded view of FIG. **33**. Depending on the size or interior volume of container **300**, its floor subassembly **302** may comprise one or more panel subassemblies **354**. In this example, four (4)

panel subassemblies **354** are provided.
(90) Referring to FIG. **33**, a panel subassembly **354** in this embodiment comprises a metal sheet **356** of suitable strength for waste storage and disposal applications and that is generally rectangular in shape. It may be formed, for example, of 7 or 12 gauge steel or of another suitable material. Flanges **358** depend perpendicularly from at least two opposite edges of sheet **356** and may be formed, for example, via a bending process. A first side **360** of sheet **356** may face an interior of container **300** in use and an opposite second side **362** of sheet **356** may face the exterior (e.g., bottom) of container **300** in use.
(91) A plurality of support members **364** are coupled with second side **362** of sheet **356** and extend between flanges **358**. In this embodiment, three (3) such support members **364** are provided, though any suitable number can be provided in other embodiments. Each support member **364** can be formed of a suitable metal material and comprise a length of channel or square tubing. In various embodiments, support members **364** can be coupled with sheet **356** by any suitable method, such as by releasable fasteners or via welding.
(92) In order that multiple panel subassemblies **354** can be coupled together to form floor assembly **302**, at least one support member **364** defines a plurality of apertures along a face thereof that extends perpendicularly to sheet **356**. Whether one or multiple support members **364** defines apertures therein can depend, for example, on the location of a particular panel subassembly **354** in the floor assembly **302**. Referring also to FIG. **32**, for instance, the leftmost and rightmost panel subassemblies **354** may each have one support member **364** defining apertures, whereas the two interior panel subassemblies **354** may have two support members **364** defining apertures.
(93) To couple first and second panel subassemblies **354** together in one example, the panel subassemblies **354** first are brought together so that the faces of support members **364** defining apertures are adjacent to one another. The apertures in the support members **364** are defined such that, when the sheet **356** of the first panel subassembly **354** is generally parallel with the sheet **356** of the second panel subassembly **354** and the flanges **358** of the first panel subassembly **354** are generally parallel to the flanges **358** of the second panel subassembly **354**, the apertures in the respective support members are in alignment. Then, suitable fasteners may be used to releasably fasten the first and second panel subassemblies **354** together. A similar approach may be used to add additional panel subassemblies **354**, as needed or desired, to a given floor assembly **302**.
(94) Turning now to FIGS. **34-35**, a still further embodiment of the present invention is shown. In this embodiment, a container **400** is a rectangular open-topped roll-of container comprising a floor assembly **402**, two side wall assemblies **404** in facing opposition, and a front wall **406**. Container **400** also can comprise either a rear wall or a door, although neither is shown in this embodiment. Container **400** in this embodiment is analogous in certain respects to container **300**, described above. For instance, floor assembly **402** in this embodiment is identical to floor assembly **302**.
(95) In this embodiment of container **400**, each side wall assembly **404** comprises a corner post **410**, a plurality of side panels **412**, and a top rail **414**. Again, the number of side panels **412** depends on the size of each side panel, the size or volume and shape of container **400**, the orientation of side panels **412**, and the particular side of the container **400** at issue. Where, as shown, a side wall assembly **404** is provided on the long sides of an open-topped roll-off container, in one embodiment three (3) such panels may be provided. In this embodiment, side panels **412** are oriented horizontally (e.g., with a longitudinal axis of each side panel **412** being parallel with an axis defined along the long dimension of container **400**), rather than vertically or at another angle. Of course, in other embodiments and for other sides of a container, different numbers and configurations of panels **412** may be used. Again, in the embodiment of container **400**, a plurality of side panels **412** also could be used for any of front wall **406**, a door, and/or floor assembly **402**.
(96) Top rails **414** are generally analogous to top rails **314** above, except in this embodiment top rails **414** comprise a length of tubing that is triangular in cross-section. Corner posts **410** are coupled with the lateral edges of several side panels **412**, for example via mechanical fasteners or

via welding, either alone or in combination with the use of locating features such as those described herein. Corner posts **410** also extend between top rails **414** and floor assembly **402**. Corner posts **410** can be coupled with floor assembly **402** via any suitable method, including mechanical fasteners as described herein or via welding, either alone or in combination with locating features such as those described above in connection with FIG. **18**. Again, in a mechanically-fastened application, various embodiments may also include a mounting bracket and/or securing gusset.

(97) Front wall **406** in this embodiment is generally analogous to front wall **306** described above, except in this embodiment its top rail **416** is similar to top rail **414**, and the lateral edges of its metal sheet **418** define a shape that corresponds to the profile of the panels **412** that form each side wall assembly **404**. For instance, here, the lateral edges of metal sheet **418** define a plurality of peaks **420** so that the edges of sheet **418** correspond to the profile of each side wall assembly **404**, which may assist with mechanical and/or welded mating. As with metal sheet **318** above, metal sheet **418** also could define one or more flanges that project perpendicularly toward the interior volume of container **400**, and such flanges could be shaped to define a surface that corresponds to an interior surface of the connected side panels **412** of side wall assembly **404**. Similarly, corner posts **410** in this embodiment are generally analogous to corner posts **310**, described above, except that the distal edges of the flanges of corner posts **410** define a shape that corresponds to the shape of the panels **412** hat form the side wall assembly **404**. Thus, here, corner posts **410** have flanges defining a plurality of peaks **422** so that the flanges mate flush with a side wall assembly **404** once assembled. As discussed above in connection with corner posts **310**, though, other variations are contemplated and within the scope of the present invention.

(98) As shown in FIGS. **34-35**, side panels **412** in this embodiment comprise a metal sheet of suitable strength for refuse disposal applications (e.g., 7 or 12 gauge steel), though other suitable materials may be used. Side panels **412** can be generally rectangular in shape, though each defines a cross section that is shaped like an arrow or “V”, with the point of the arrow or V facing toward the exterior of the container **400**. Side panels **412** can be formed from a single unitary sheet, for instance by a bending process, though this is not required in all embodiments.

(99) In general, and as with side panels **312** described above, each side panel **412** can define a plurality of apertures at various positions around its perimeter that, during assembly, can be aligned with correspondingly-defined apertures in other side panel(s) **412**, corner posts **410**, and/or top rail **414** (not shown, but analogous to those described above in connection with container **100**) to releasably fasten these various components to one another. As those of skill in the art will appreciate, the number and placement of apertures in a given side panel **412** will depend on its location within a side wall assembly **404**, and each side panel **412** may not have the same number and/or configuration of apertures as the other side panels **412**. Thus, apertures in side panel **412** are not shown in this embodiment, but they can be analogous to the apertures **152** of side panels **114** shown above, and the apertures in corner posts **410** and/or top rail **414** can be analogous to apertures **153**, described above. In various embodiments, suitable fasteners include, but are not limited to, bolts screws, nails, and other mechanical fasteners such as those described in connection with container **100** above, and of course the type, number, and grade of fasteners used will vary, as needed or desired, in various embodiments.

(100) In various embodiments, side panels **412** can be coupled to the top rail **414**, to each other, to front wall **406**, to corner post **410**, and to floor assembly **402** in any of the manners described herein. Thus, in certain embodiments, mechanical fasteners are used, either alone or in combination with locating features. In various embodiments, welding can be used in the alternative or in addition, as can mounting brackets and/or securing gussets.

(101) Based on the foregoing, it will be appreciated that embodiments of the invention provide improved containers and modular wall assemblies therefor. Many modifications and other embodiments of the inventions set forth herein will come to mind to one skilled in the art to which these inventions pertain having the benefit of the teachings presented in the foregoing descriptions

and the associated drawings. Therefore, it is to be understood that the inventions are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Moreover, although the foregoing descriptions and the associated drawings describe exemplary embodiments in the context of certain exemplary combinations of elements and/or functions, it should be appreciated that different combinations of elements and/or functions may be provided by alternative embodiments without departing from the scope of the appended claims. In this regard, for example, different combinations of elements and/or functions than those explicitly described above are also contemplated as may be set forth in some of the appended claims. In cases where advantages, benefits or solutions to problems are described herein, it should be appreciated that such advantages, benefits and/or solutions may be applicable to some example embodiments, but not necessarily all example embodiments. Thus, any advantages, benefits or solutions described herein should not be thought of as being critical, required or essential to all embodiments or to that which is claimed herein. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A container, comprising: a first side wall, a second side wall opposite the first side wall, a front wall coupled with the first and second side walls, a rear wall opposite the front wall and coupled with at least one of the first and second side walls, and a bottom wall coupled with at least the first and second side walls and the front wall; the first side wall, second side wall, front wall, rear wall, and bottom wall together defining an open-topped interior volume of at least ten (10) cubic yards; at least one of the first side wall, second side wall, front wall, and rear wall comprising: a top rail; and a plurality of side panels; the bottom wall comprising a floor assembly, the floor assembly comprising at least two panel subassemblies releasably fastened together, each panel subassembly comprising a metal sheet having a first side facing the interior volume and a second side opposite the first side, wherein a plurality of support members are coupled with the second side of the metal sheet; and wherein the plurality of support members extend between flanges depending from opposite sides of the metal sheet, wherein the flanges are formed from bending a portion of the metal sheet.
2. The container of claim 1, wherein a first panel subassembly of the at least two panel subassemblies is disposed with respect to a second panel subassembly of the at least two panel subassemblies such that one of the plurality of support members of the first panel subassembly is in engagement with one of the plurality of support members of the second panel subassembly.
3. The container of claim 2, wherein the support member of the first panel subassembly defines a first set of apertures that are aligned with a second set of apertures defined in the support member of the second panel subassembly, and wherein a plurality of mechanical fasteners extend through the first and second sets of apertures to fasten the first panel subassembly and the second panel subassembly together.
4. The container of claim 1, wherein the at least two panel subassemblies comprises four panel subassemblies.
5. The container of claim 1, wherein each of the support members comprises a length of channel or square tubing.
6. The container of claim 1, wherein the plurality of support members includes three or more support members, and two support members of the three or more support members are located at opposite lateral edges of the sheet.
7. A container, comprising: first, second, and third side walls and an open end; a bottom wall coupled with the first, second, and third side walls; a door pivotably coupled to the container opposite the third side wall and movable between a first position at which the door is spaced apart

from the at least one open end and a second position at which the door closes the at least one open end; wherein the container defines an open-topped interior volume of at least ten (10) cubic yards; wherein at least one of the first, second, and third side walls comprises a plurality of rectangular side panels; and wherein the bottom wall comprises a floor subassembly, the floor subassembly comprising first, second, and third panel subassemblies each comprising a sheet supported by one or more support members, wherein the first panel subassembly comprises a first support member in facing opposition to a second support member of the second panel subassembly, wherein the second panel subassembly comprises a third support member in facing opposition to a fourth support member of the third panel subassembly, wherein the first, second, third, and fourth support members extend perpendicularly to the first and second side walls, wherein the sheets of the first, second, and third panel subassemblies each define a pair of laterally opposed depending flanges formed by bending portions of the respective sheets, and wherein the first, second, third, and fourth support members each extend between the pair of laterally opposed depending flanges.
8. The container of claim 7, wherein the first, second, and third panel subassemblies are releasably fastened together by mechanical fasteners.
9. The container of claim 7, wherein the one or more support members includes three or more support members, and two support members of the three or more support members are located at opposite lateral edges of the sheet.
10. A container, comprising: vertical side walls and a horizontal bottom wall coupled with the vertical side walls; the vertical side walls and horizontal bottom wall together defining an open-topped interior volume of at least ten (10) cubic yards; at least one of the vertical side walls comprising: a plurality of side panels; and a top rail fastened together with the upper ends of at least one of the plurality of side panels; the bottom wall comprising a floor subassembly, the floor subassembly comprising a plurality of panel subassemblies, each of the plurality of panel subassemblies comprising: a metal sheet having a first side facing the interior volume and an opposite second side; and at least one support member coupled with the second side of the metal sheet; wherein the plurality of panel subassemblies are brought together so that faces of support members are adjacent to one another and the adjacent support members are coupled together via mechanical fasteners.
11. The container of claim 10, wherein each of the plurality of panel subassemblies comprises a pair of depending flanges, wherein the depending flanges are formed from bending portions of the respective metal sheet.
12. The container of claim 10, wherein a first vertical side wall of the vertical side walls comprises a unitary metal sheet.
13. The container of claim 10, wherein each panel subassembly comprises a plurality of support members formed from a length of channel or tubing.
14. The container of claim 10, wherein the plurality of panel subassemblies includes three or more panel subassemblies.
15. The container of claim 10, wherein the sheet of each of the plurality of panel subassemblies extends between two laterally opposed vertical sidewalls.
16. The container of claim 10, wherein the at least one support member of each respective panel subassembly defines a face extending perpendicular to the sheet of each respective panel subassembly.
17. The container of claim 16, wherein the face of the at least one support member of each panel subassembly is in engagement with the face of the at least one support member on another panel subassembly.
18. The container of claim 10, wherein each of the plurality of panel subassemblies comprises at least two support members, the at least two support members located at opposite lateral edges of the sheet.
19. The container of claim 10, wherein the at least one support member includes the three or more

support members, and two support members of the three or more support members are located at opposite lateral edges of the sheet.